

# MARCOS PAULO MAGALHAES DA SILVA

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## PROFILE SUMMARY

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- **Senior Full Stack software engineer** with **10+ years** of experience building **AI-driven SaaS, fintech,** and **e-commerce** platforms.
- Specialized in **JavaScript/TypeScript (Node.js, NestJS, Express)** and **React**, with **cloud-native microservices** on **Azure, GCP,** and **AWS**, and hands-on experience integrating **machine learning** and **NLP** models into production systems.
- Experienced in designing **scalable backend architectures, AI-powered data pipelines,** and **responsive frontend applications** for **high-traffic, data-intensive** products.

## EDUCATION

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**National University of Singapore (NUS)**

Sep 2013 – Jan 2017

*Bachelor of Computer Science*

## PROFESSIONAL EXPERIENCE

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**Senior Software Engineer**

Sep 2023 – Present

*Business Valuation Advisors, Inc.*

*Vienna, VA, USA*

- Architected a **Node.js/TypeScript**-based valuation platform using **NestJS, PostgreSQL,** and **Azure App Service**, automating **DCF** and comparable company models and reducing manual analysis time by **45%**.
- Built **AI-powered** financial data ingestion pipelines using **Node.js services** and **Azure Cognitive Services**, extracting data from financial statements and improving input accuracy by **30%**.
- Designed scalable **microservices** on **Azure (App Service, Azure Functions, Azure SQL)** to process **10K+** valuation scenarios, maintaining **sub-second** API response times.
- Implemented **machine learning** forecasting models using **Azure Machine Learning**, improving revenue projection accuracy by **18%** across client portfolios.
- Optimized heavy financial computations by introducing enhanced **TypeScript processing** and **Azure Cache for Redis**, reducing report generation time from **12 minutes** to under **4 minutes**.
- Developed secure client dashboards with **React** and **TypeScript**, backed by **Azure AD** authentication, reducing analyst turnaround time by **35%**.
- Resolved slow, data-heavy UI rendering in financial dashboards by refactoring components with **React, TypeScript,** and **memorization strategies**, reducing page load time from **3.2s** to **1.4s** for large valuation datasets.
- Improved maintainability and consistency across the frontend by introducing a modular component architecture with **React, Tailwind,** and shared hooks, eliminating duplicated logic and reducing UI-related bugs by **35%**.
- Implemented **RBAC, audit logging,** and **encrypted storage** using **Azure Key Vault** and **Azure Monitor**, strengthening compliance for financial data handling.
- Established automated **CI/CD pipelines** with **Docker, Azure DevOps,** and **Azure Container Registry**, cutting deployment time by **60%** and enabling **weekly** production releases.

**Full Stack Engineer**

Feb 2021 – Sep 2023

*GFT Technologies*

*Barueri, Brazil*

- Rebuilt the product catalog and pricing services using **Node.js (NestJS/Express)** on **GCP (Cloud Run, Cloud SQL)** to handle flash-sale spikes, increasing catalog throughput by **2.8×** and preventing stock inconsistencies during peak campaigns.
- Solved slow and irrelevant search results by implementing an **AI-powered** product ranking pipeline using **TypeScript**-based ML services and **BigQuery ML**, improving search conversion by **21%** and reducing bounce rate by **18%**.
- Addressed checkout failures under high traffic by refactoring order and payment APIs and introducing **Redis caching** and **GCP Pub/Sub** for asynchronous processing, reducing checkout errors by **35%** during promotional events.
- Built a real-time inventory synchronization system with **GCP Pub/Sub, Cloud Functions,** and **Cloud SQL**, reducing overselling incidents by **40%** across multiple warehouses.

- Optimized recommendation services by deploying batch and real-time **ML pipelines** using **Node.js data processing services** and **Vertex AI**, increasing average order value by **16%**.
- Reduced page load times by redesigning high-traffic APIs and adding **Cloud CDN** and query optimizations, cutting **p95 response time** from **1.1s** to **520ms** during peak shopping periods.
- Automated order processing and logistics events using **Node.js background workers** on **Cloud Run** and **Pub/Sub**, increasing fulfillment processing speed by **2.3×** during seasonal campaigns.
- Enhanced release reliability by containerizing services with **Docker** and setting up **GCP CI/CD pipelines** using **Cloud Build** and **Artifact Registry**, reducing deployment failures by **55%** and shortening release cycles from **2 hours** to **25 minutes**.

## Mid Full Stack Developer

Sep 2018 – Feb 2021

*Freshworks*

*The Metropolis, Singapore*

- Re-architected the ticket ingestion pipeline into **Node.js microservices** using **TypeScript**, background workers, and **Redis**, processing **1.2M** support events/day, which increased throughput by **3×** and cut queue backlogs by **40%** during peak hours.
- Built an **NLP classification** service using **ML pipelines** to auto-tag incoming tickets across **30+** categories, improving tagging accuracy from **68%** to **85%** and reducing manual triage time by **35%**.
- Streamlined high-traffic customer and ticket APIs by refactoring **PostgreSQL** queries, adding **Redis caching**, and introducing indexed search fields, reducing **p95 response time** from **820 ms** to **450 ms** under peak load.
- Removed model inference bottlenecks by introducing **batch prediction workers** and lightweight inference endpoints, reducing average prediction latency from **1.2 s** to **600 ms** and lowering compute costs by **20%**.
- Stabilized releases by containerizing services with **Docker**, setting up **AWS CodePipeline + EC2** deployments, and adding systematized test gates, cutting production deployment failures by **60%** and reducing release time from **3 hours** to **18 minutes**.

## Junior Software Developer

Feb 2017 – Sep 2018

*Ninja Van*

*Singapore, Singapore*

- Built backend features and **REST APIs** using **Node.js** and **Express**, with database tuning and standardized validation.
- Wrote unit and integration tests using **Jest**.
- Containerized services with **Docker** and supported **AWS** deployments in Agile teams.

## Software Developer (Intern)

Feb 2016 – Dec 2016

*Grab*

*Singapore, Singapore*

- Developed backend features using **Node.js REST services**, working with relational databases (**SQL**).
- Wrote unit tests and supported build pipelines under senior supervision.
- Built UI components using **HTML**, **CSS**, **JavaScript**, and **jQuery**, integrating frontend views with backend APIs.

## PROJECTS

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### Business Valuation and Analysis Application

- Financial valuation platform using **Node.js (NestJS)**, **Azure SQL**, and **Azure App Service**, with **QuickBooks API** integration to automatically ingest P&L, balance sheets, and transaction data. Added **AI-assisted** data extraction and normalization, reducing analyst data preparation time by **40%** and cutting full valuation turnaround from **12 minutes** to under **4 minutes**.

### E-commerce Platform Optimization and AI-Driven Personalization

- **GCP-based** e-commerce platform using **Node.js (NestJS/Express)**, **Pub/Sub**, and **Vertex AI**, delivering scalable catalog, checkout, and recommendation systems that boosted search conversion by **21%**, escalated order value by **16%**, and decreased checkout errors by **35%** during peak campaigns.

### Powered Customer Support Automation Platform

- **Node.js microservices** platform reduced with **TypeScript**, **Redis**, and **PostgreSQL** to process **1.2M** support events/day, delivering **NLP-driven** ticket classification and streamlined APIs that increased throughput by **3×**, boosted tagging accuracy to **85%**, and reduced **p95 latency** by **45%**.

## TECHNICAL SKILLS

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**Backend Development:** Node.js, TypeScript, NestJS, Express, REST APIs, microservices architecture

**Frontend Development:** React, TypeScript, JavaScript, HTML, CSS, Tailwind

**AI & Data:** ML service integrations, data processing pipelines, model inference services

**Cloud & DevOps:** Azure, GCP, AWS, Docker, CI/CD pipelines, Cloud Run, App Service, Azure Functions

**Databases & Caching:** PostgreSQL, Cloud SQL, Azure SQL, Redis, BigQuery

**Messaging & Async Processing:** Node.js workers, GCP Pub/Sub, background workers, task queues

**Testing & Tooling:** Jest, unit testing, integration testing, Git, Agile/Scrum